



ART: Automated Redox-Titration with Interdigitated Electrode Arrays for Energy Storage Materials



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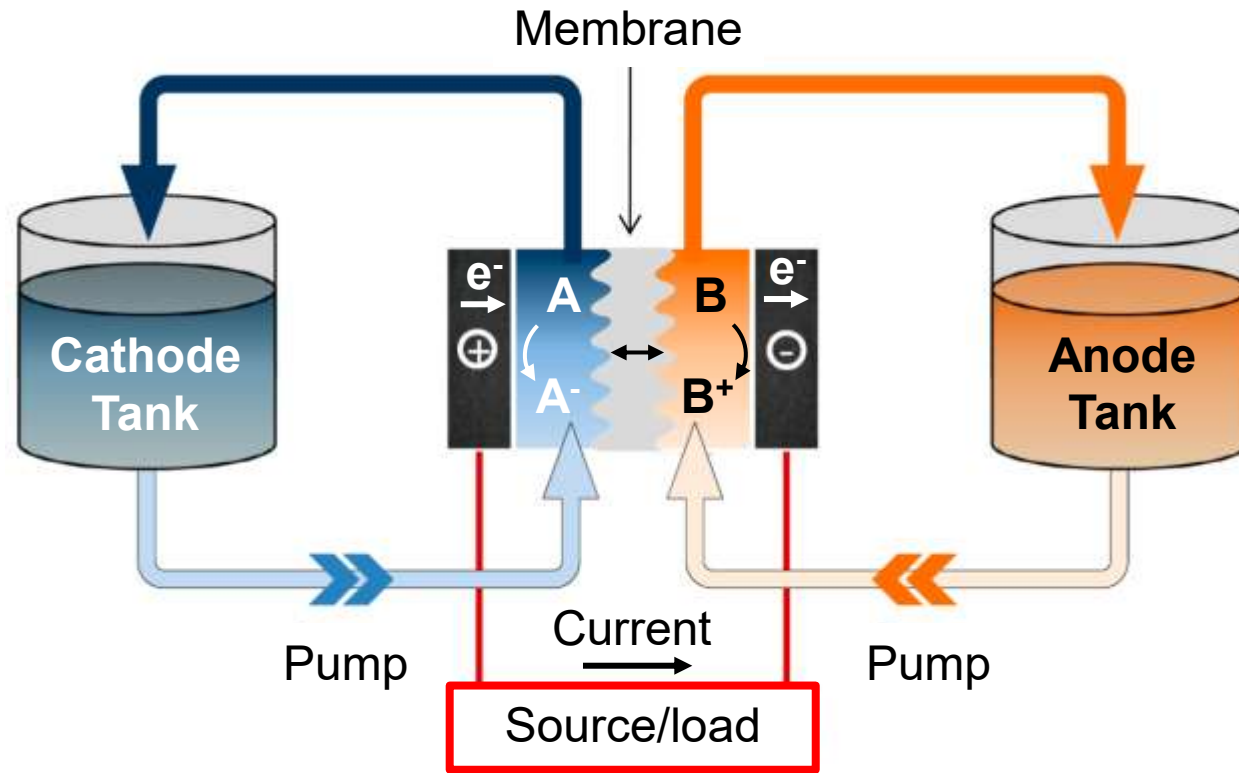
247th ECS Meeting, Montreal, Canada

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Redox Flow Batteries for Grid-Level Energy Storage

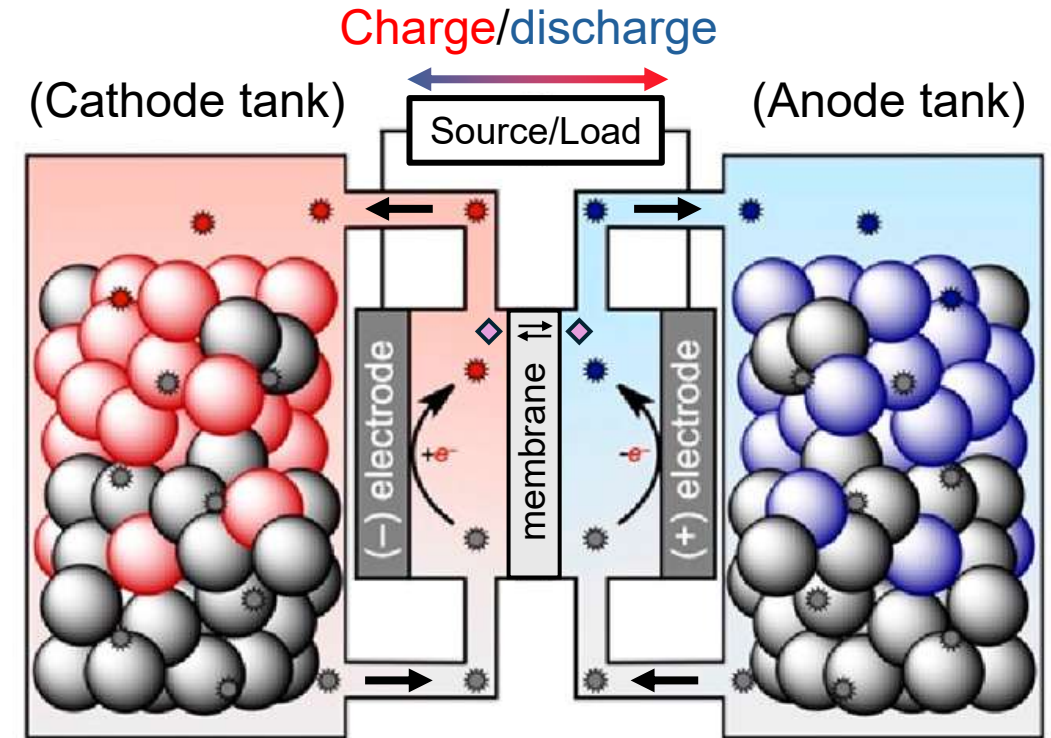


Redox flow battery (RFB)



- Highly scalable, decoupled power and capacity
- **Limitation:** Energy density is limited by the solubility of redox-active species

Redox targeting flow battery (RTFB)

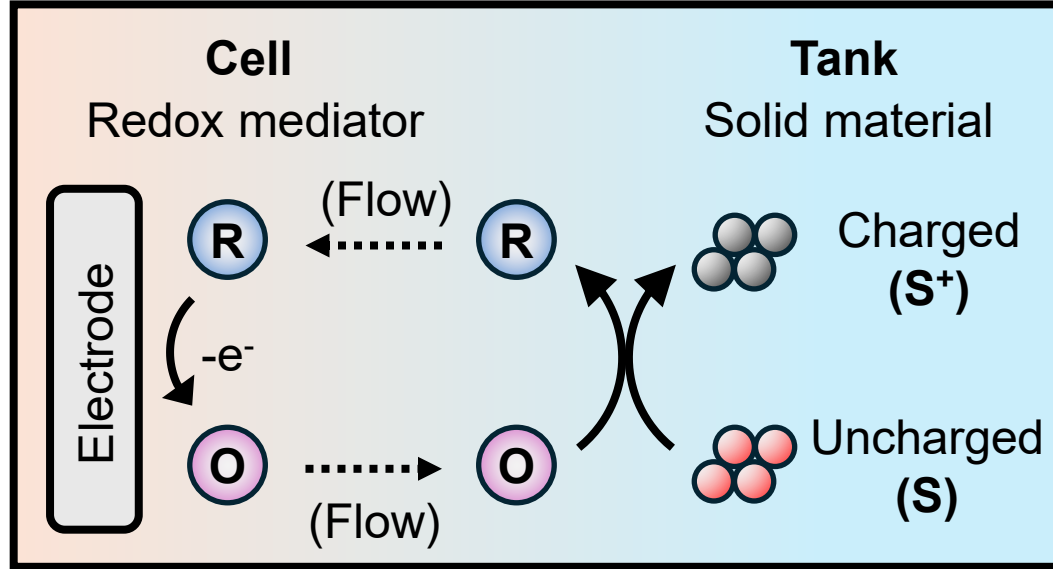


- Stores energy in a solid material
 - Active species concentration up to **20 M**
- Redox mediators shuttle charges between the solid and electrode via a mediated electron transfer process known as **redox targeting**

The Redox Targeting Process in RTFBs



The redox targeting process

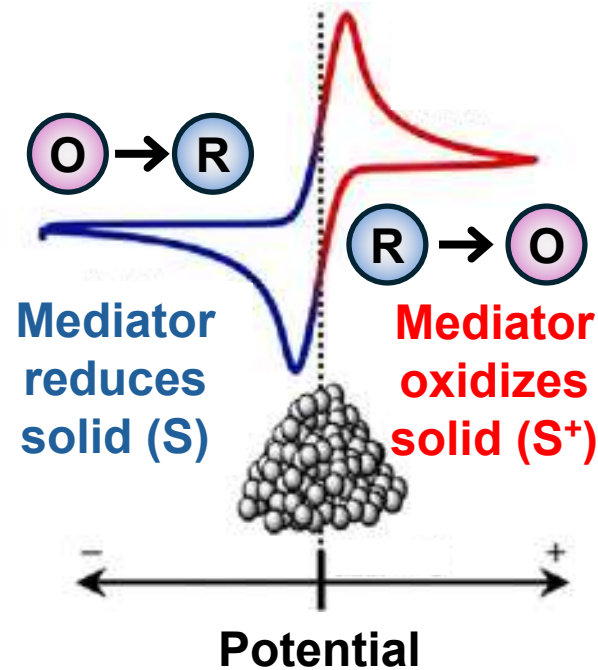


- Active redox mediator generation in the cell
- Solid material chemically charged/discharged in the tank
- **Thermodynamic driving force:** Nernstian potential difference
- **Kinetics:** Follows the Butler-Volmer model

$$k_{red} = k^0 e^{-\alpha f \Delta E}, k_{ox} = k^0 e^{(1-\alpha) f \Delta E}$$

Single mediator redox targeting

$$E_{O/R}^{0'} = E_{solid}^{0'}$$

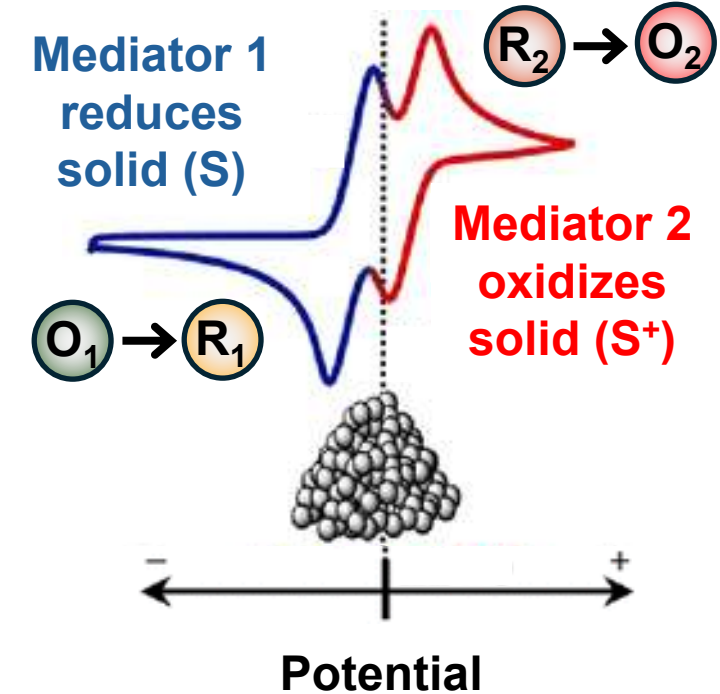


Pros: Simple
Cons: Slow kinetics

Dual mediator redox targeting

$$E_{O_1/R_1}^{0'} < E_{solid}^{0'}$$

$$E_{O_2/R_2}^{0'} > E_{solid}^{0'}$$



Pros: Fast kinetics
Cons: Complex, low voltage efficiency

Challenges in Studying Redox Targeting



Huge chemical/material space to study Solid material

- Inorganic and organic materials

Redox mediators

- Mediators with different $E^{o'}$

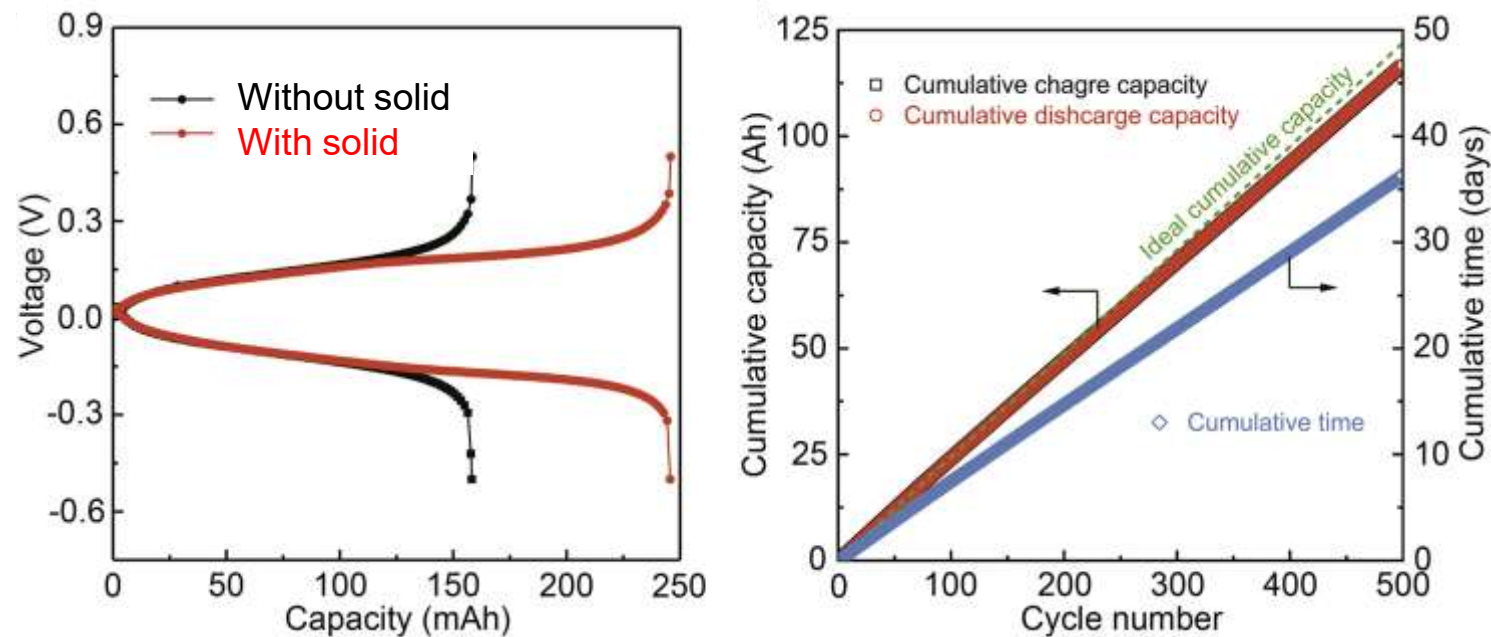
Electrolyte compositions

- Charge-balancing ions
- Electrolyte concentration

Figure of merits

- Fast redox targeting kinetics
- Large capacity for the solid material

Battery cycling experiments



Summary

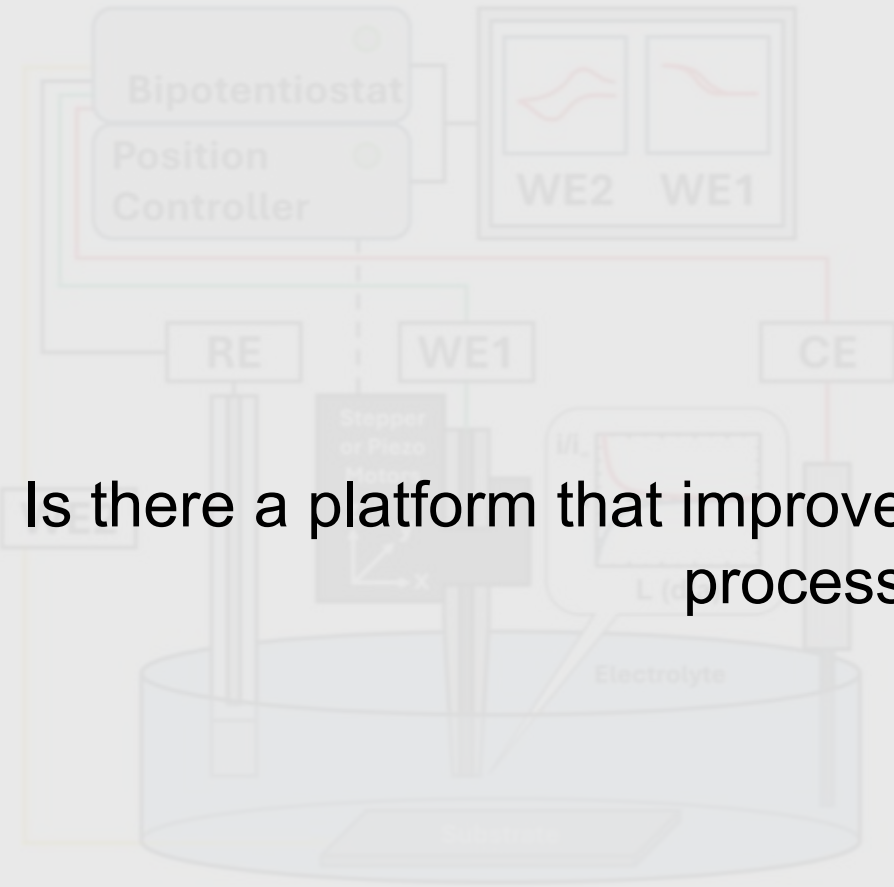
Pros: Offers device-level metrics

Cons: Time-consuming, does not directly offer mechanistic insights about the redox targeting process

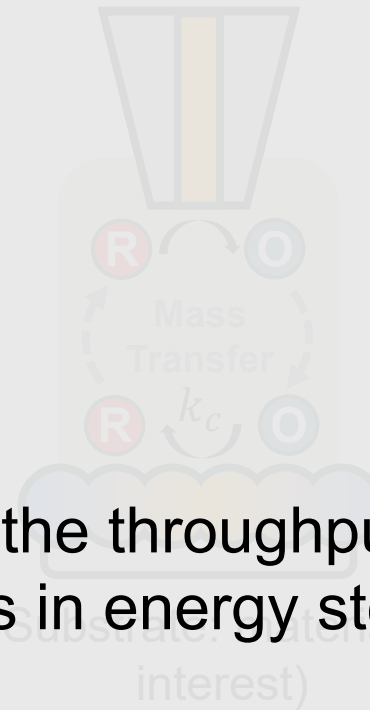
Is there a platform that provides more mechanistic insights?



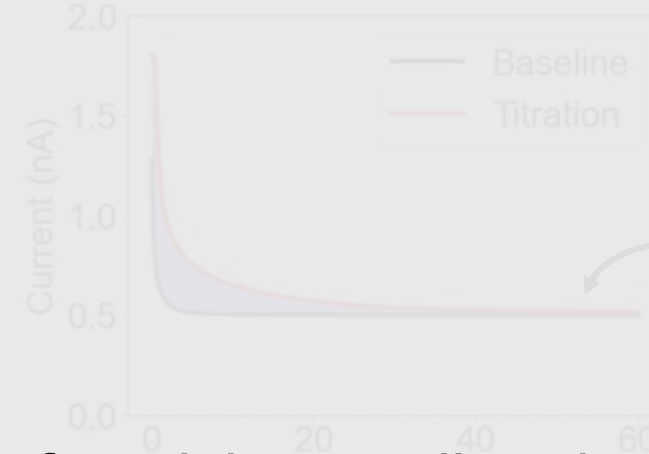
SECM schematics



(Tip: Ultramicroelectrode)



Current vs. time response

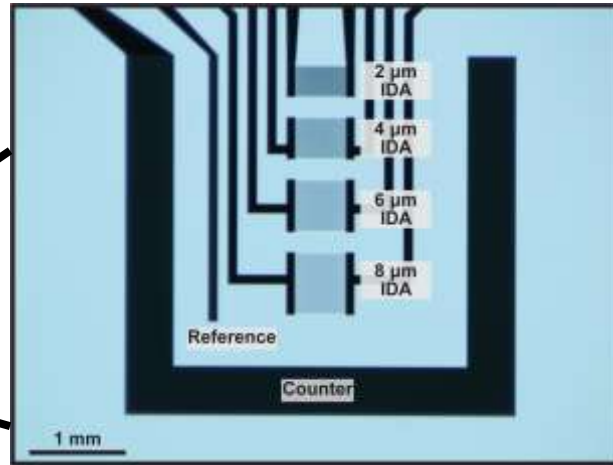
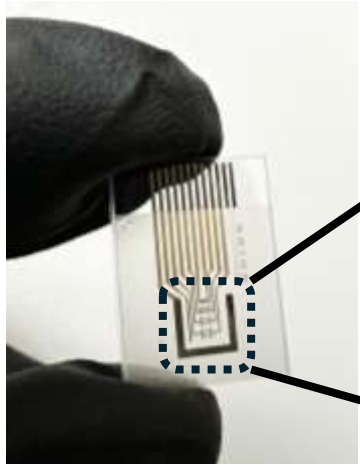


Is there a platform that improves the throughput of studying mediated electron transfer processes in energy storage materials?

Redox titration with SECM

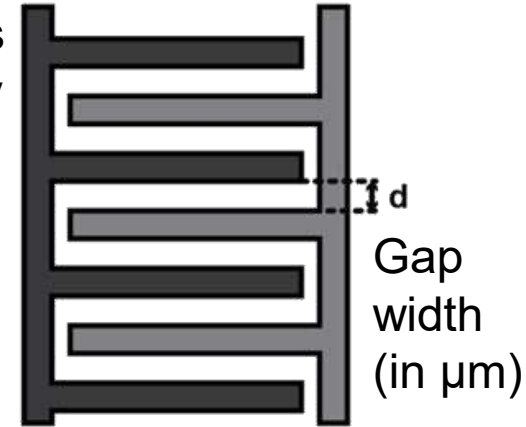
- Tip generates active species to titrate the charges in the substrate
- A transient current increase can be observed
- Can couple with simulations to understand the kinetics
- **Low throughput, not ideal for studying redox targeting due to the large chemical/material space to explore**

Interdigitated Electrode Arrays (IDAs)

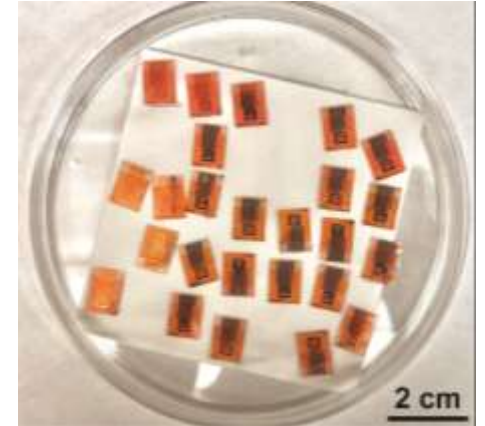


4 IDAs with different gap widths per chip

25 bands per array



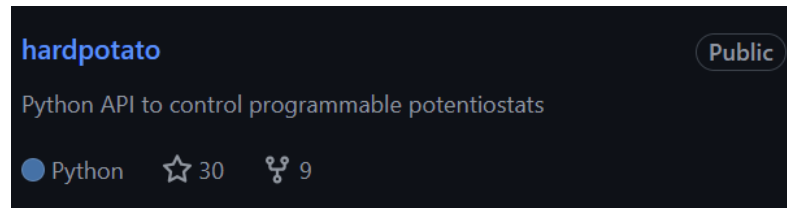
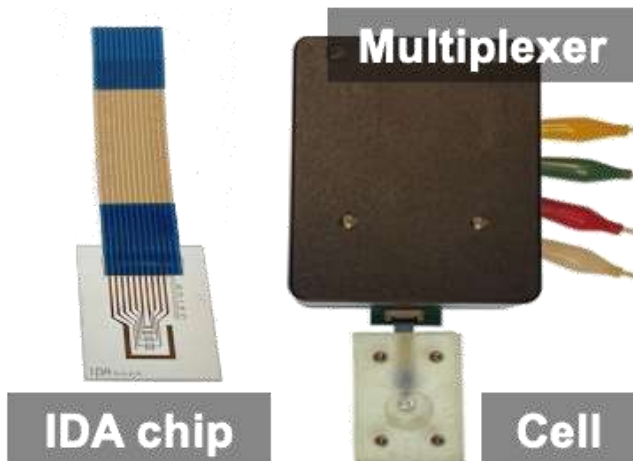
Two **individually addressable** working electrode arrays



Batched, high-yield microfabrication

Multiplexing circuit to control IDAs

Python library to control commercial electrochemical workstations



<https://github.com/jrLAB/hardpotato>
ACS Meas. Sci. Au **2023**, 3 (1), 62–72
Anal. Chem. **2023**, 95 (11), 4840–4845
Device **2023**, 1 (5), 100103–100103

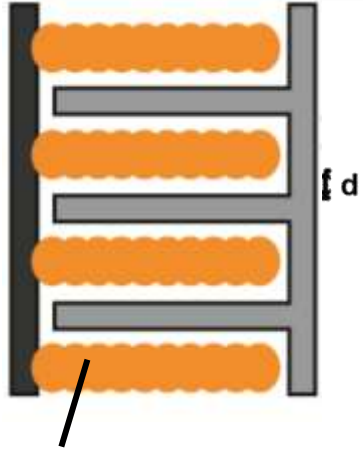
Summary

- **Automated** electrochemical measurements can be performed on IDAs using Python
- **Large, combinatorial studies** can be done

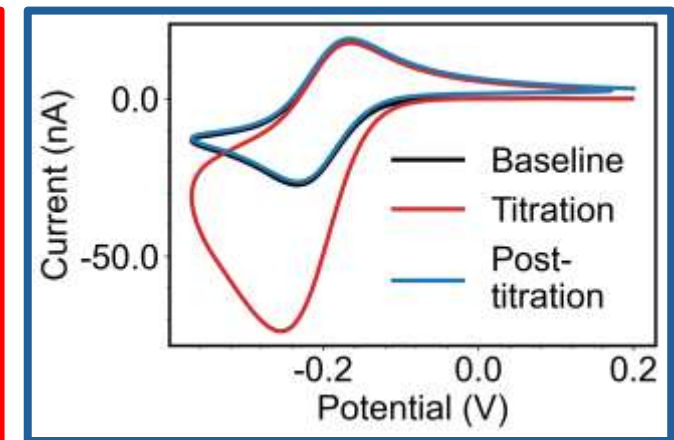
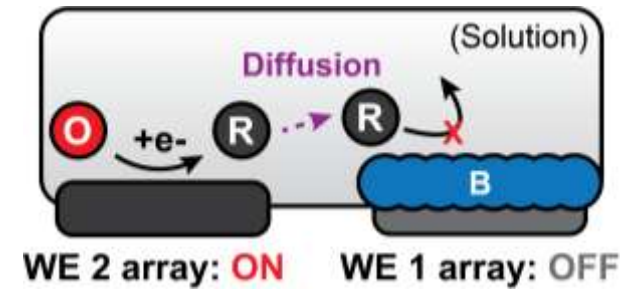
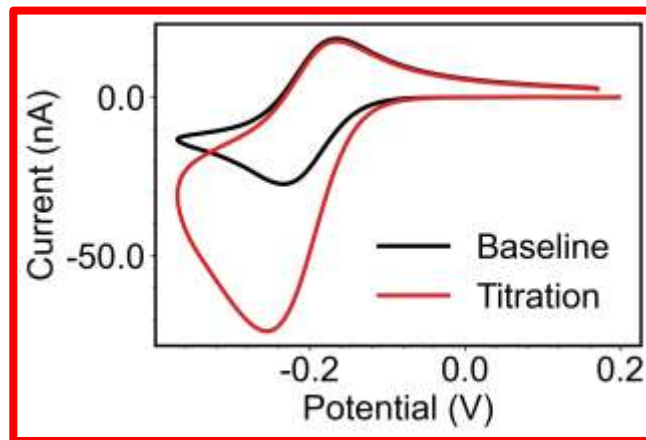
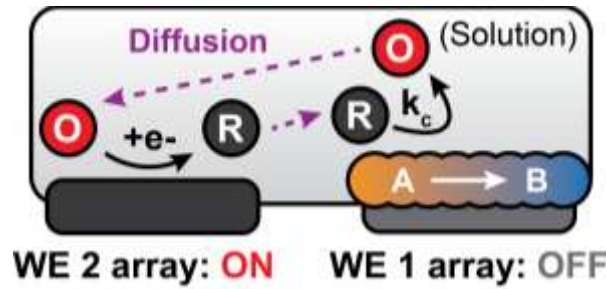
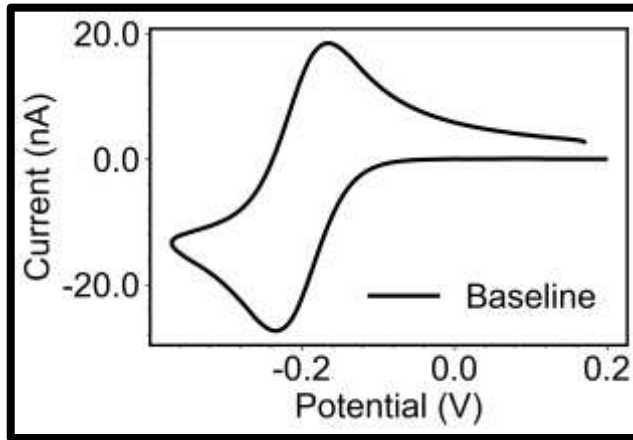
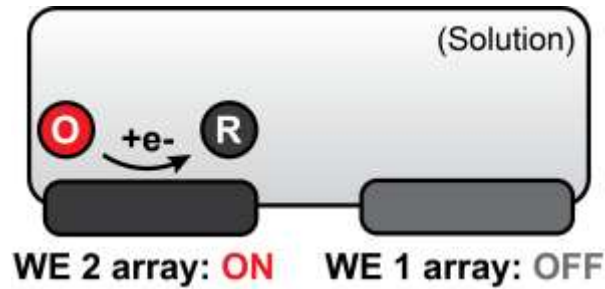
Redox-Titration with IDAs



Array 1 Array 2



Material of
interest

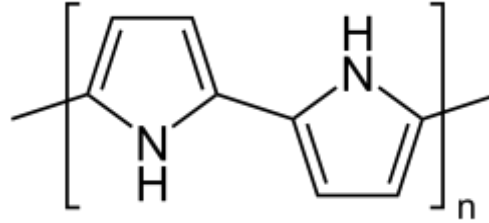
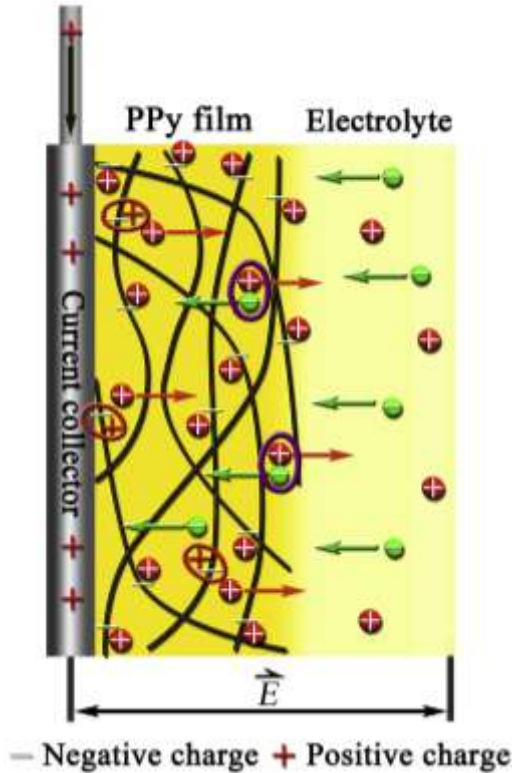


- Performing a cyclic voltammetry (CV) with array 2 alone generates a normal “duck-shape” CV
- Pattern the material of interest on array 1
- A **transient current increase** will be observed on the redox titration CV
- Current level returns to the baseline after the material has been fully titrated

Redox Titration of Polypyrrole, a Model System



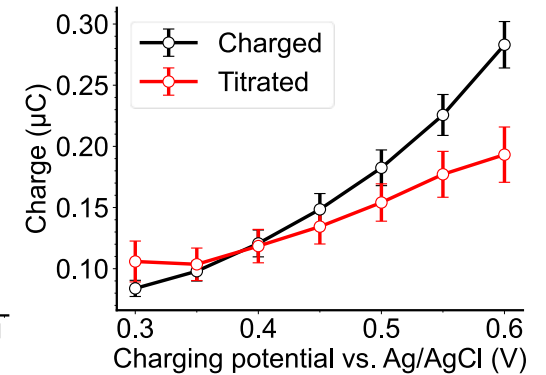
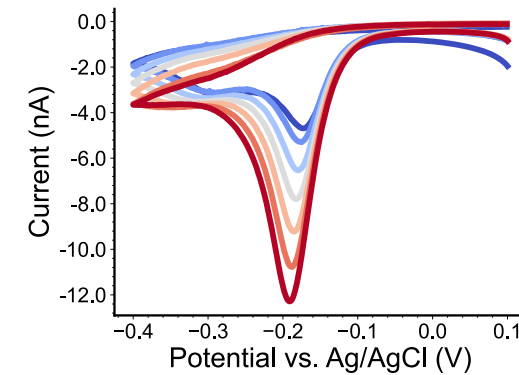
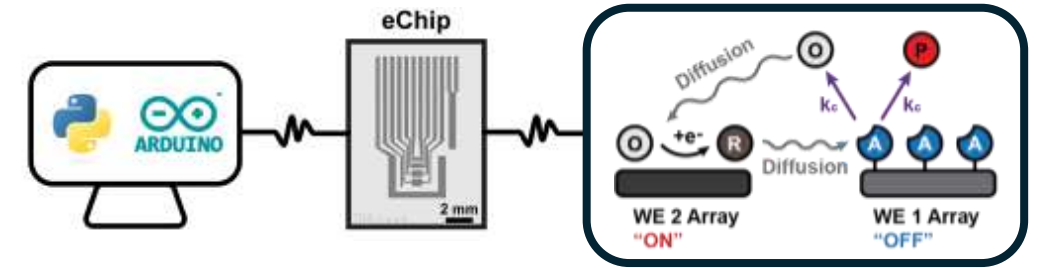
- Polypyrrole (PPy) is a conjugated polymer
- Candidate material for electrochemical supercapacitors and dual-ion batteries



Motivation

- Polypyrrole can store anions from the electrolyte once oxidized
- Use automated redox titration to study the charge storage behavior of PPy under different conditions

Automated Redox Titration of Polypyrrole, a Conductive Polymer

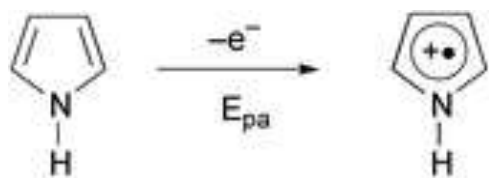


Wang, Z; Pence, M; Rodríguez-López, J.
(Manuscript in preparation)

Electrodeposition of Polypyrrole (PPy) on IDAs



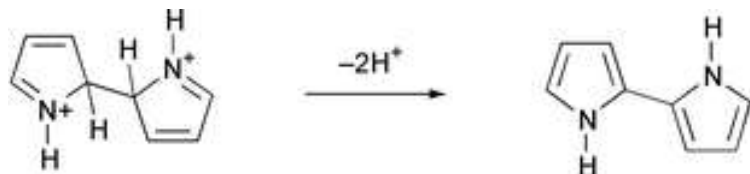
1. Monomer oxidation



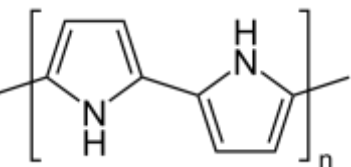
2. Dimerization



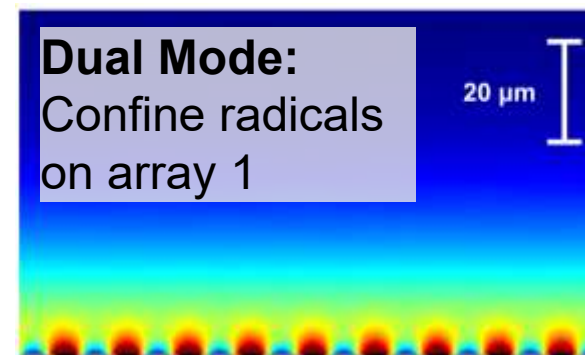
3. Stabilization



4. Additional combination-stabilizations



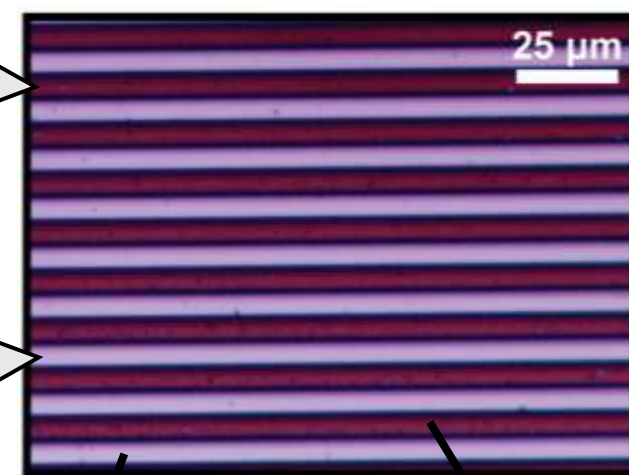
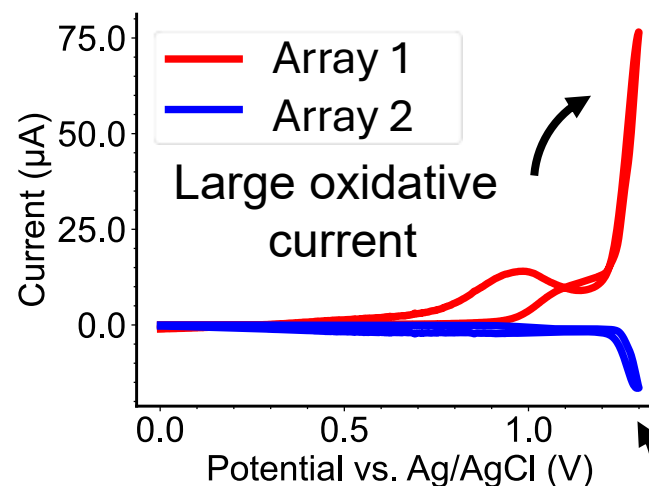
- PPy must be selectively patterned on one of the electrode arrays
- Use **dual-mode** to limit the monomeric radicals on one of the arrays



Array 2
ON:
Reduction

Array 1
ON:
Oxidation

Dual mode electrodeposition CV Optical micrograph **after** deposition



Small reducing current

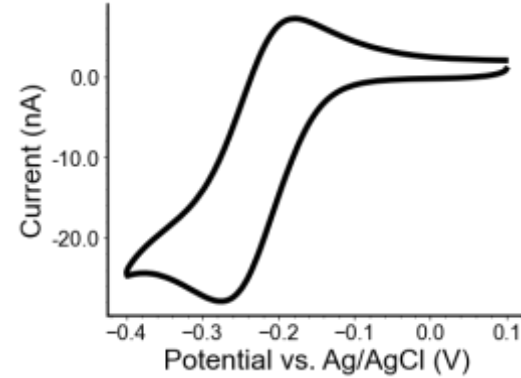
Array 2 (Clean) Array 1 (with PPy)

WE: Pt, RE: Ag/AgCl, CE: graphite. WE 2 potential: 0 V vs. Ag/AgCl.
10 mM pyrrole, 0.1 M TBAPF₆ in ACN; scan rate: 500 mV/s

Experimental Workflow and Response



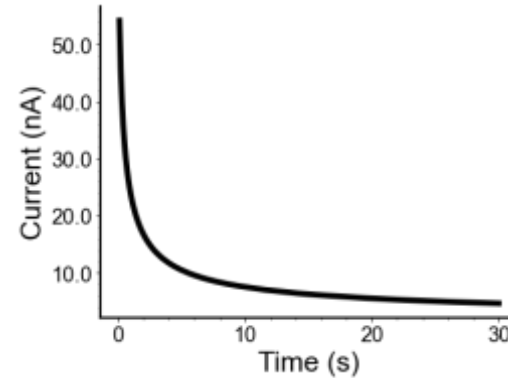
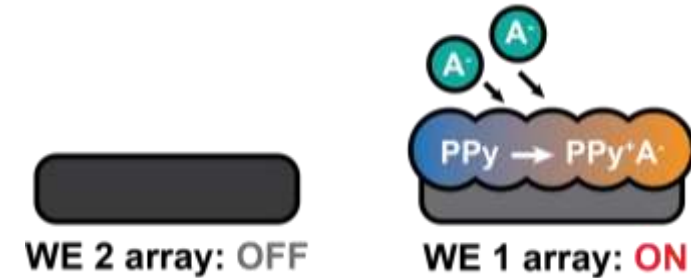
1. Baseline CV with array 2



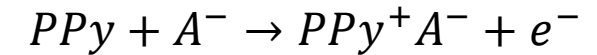
WE: Pt, RE: Ag/AgCl, CE: graphite.
0.2 mM $\text{Ru}(\text{NH}_3)_6^{3+}$, 0.1/0.5 M KCl, scan rate: 10/25/50 mV/s

- A “duck-shape” baseline CV is obtained

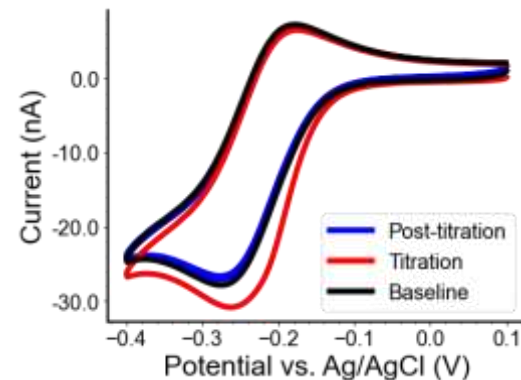
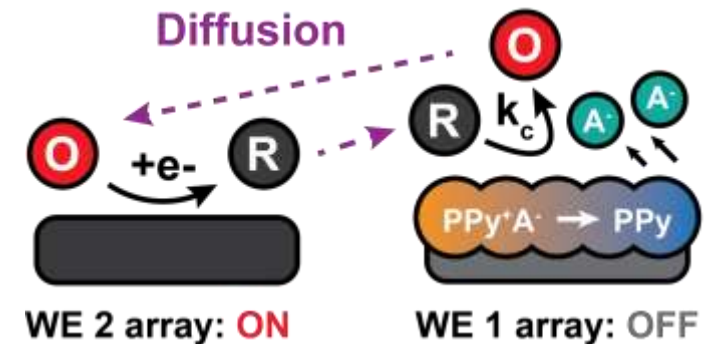
2. PPy charging with array 1



- An oxidizing potential is applied
- PPy gets oxidized and incorporates anions from electrolyte



3. Titration and post-titration CVs with array 2

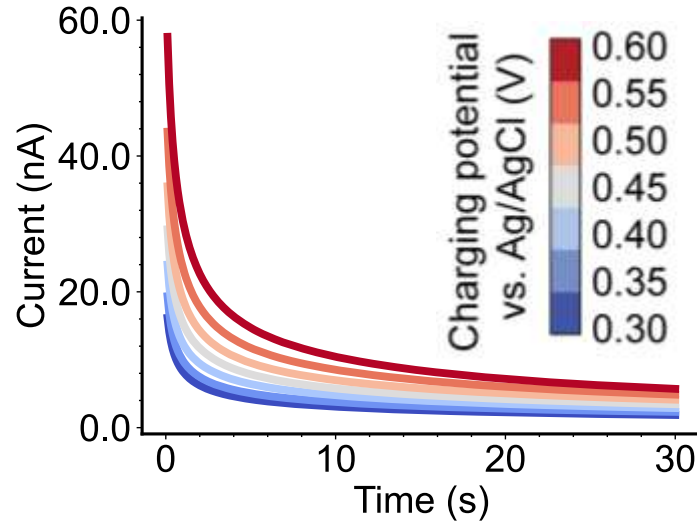


- Same CV as the baseline was conducted
- A total of 3 cycles:
 - First cycle: Redox titration CV
 - Following cycles: Post-titration validation
- Post-titration CV current returns to the baseline level

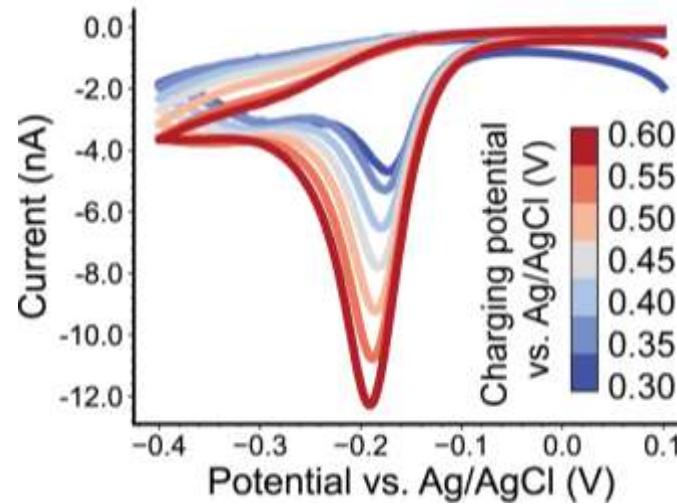
Behavior of PPy under Low Ionic Strength (0.1 M KCl)



PPy charging curves at different potentials



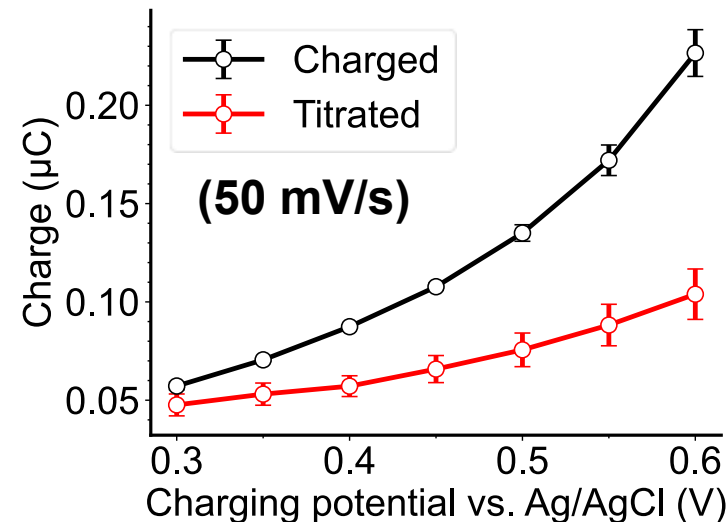
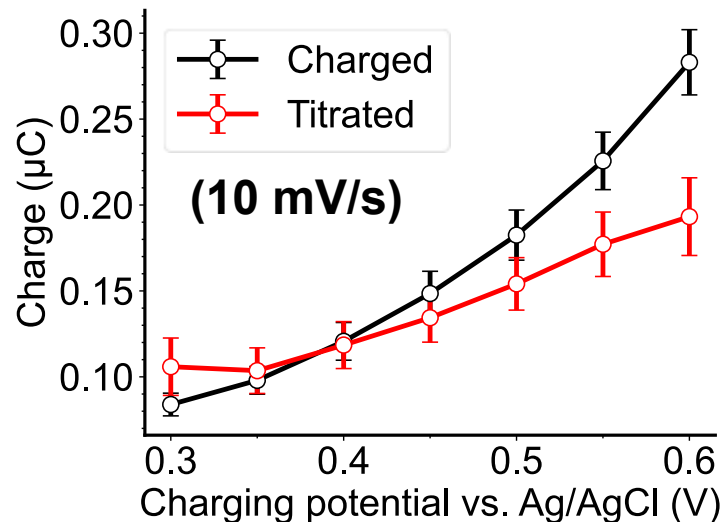
Baseline-subtracted titration CVs (10 mV/s scan rate)



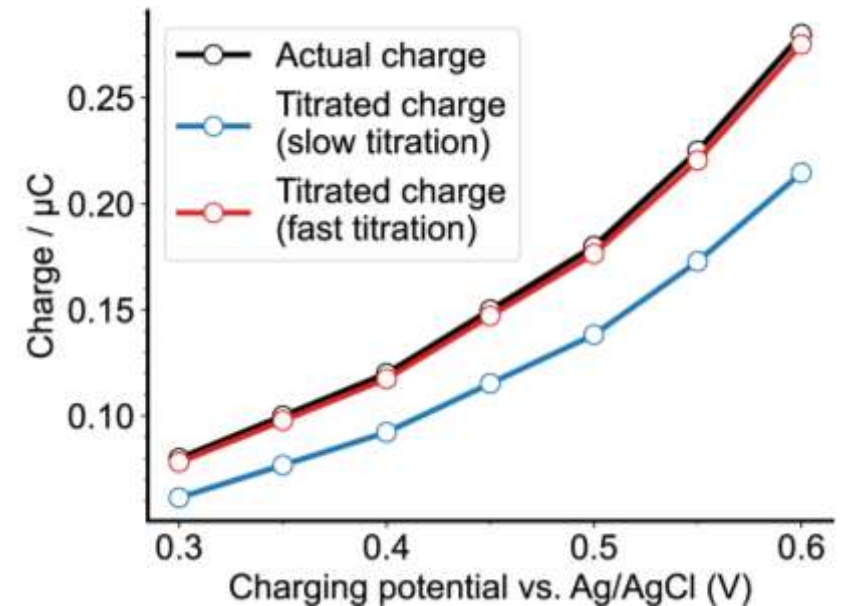
* Data shown here were from the IDA with 2 μm gap width

- Integrate to calculate the stored charge
- Integrate to obtain the titrated charge
- Finite-element simulations via COMSOL reveals the role of redox titration kinetics
 - Slow kinetics
 - Cannot establish full titration within the timescale of the CV measurement
 - Less titrated charge at a higher CV scan rate

Stored vs. titrated charge comparison



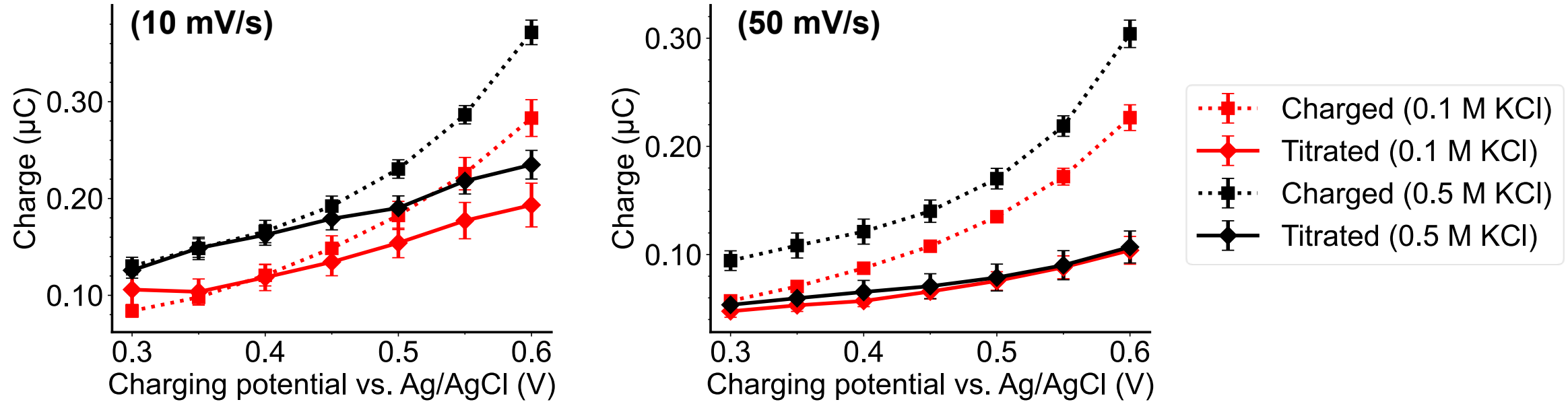
Simulated behavior (10 mV/s)



Redox Titration under Elevated Ionic Strength (0.5 M KCl)



Stored vs. titrated charge comparison

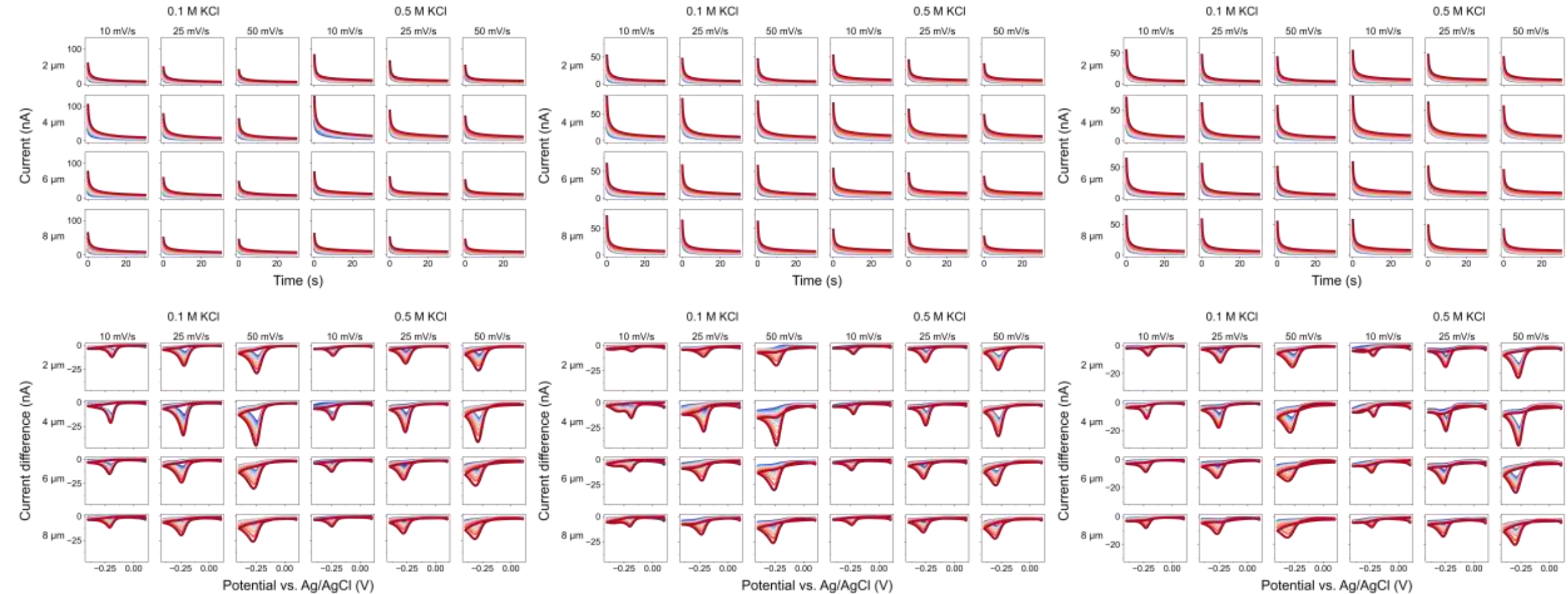
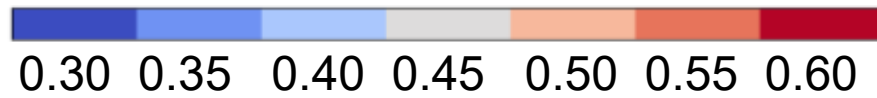


- Elevated ionic strength (**0.5 M KCl**) leads to more charge storage in PPy
- The influence of electrolyte concentration on the amount of titrated charge only shows up at low CV scan rate
 - **Further confirms the slow redox-titration kinetics suggested by simulation**

High-Throughput Nature of Our Platform



PPy charging potential (vs. Ag/AgCl)



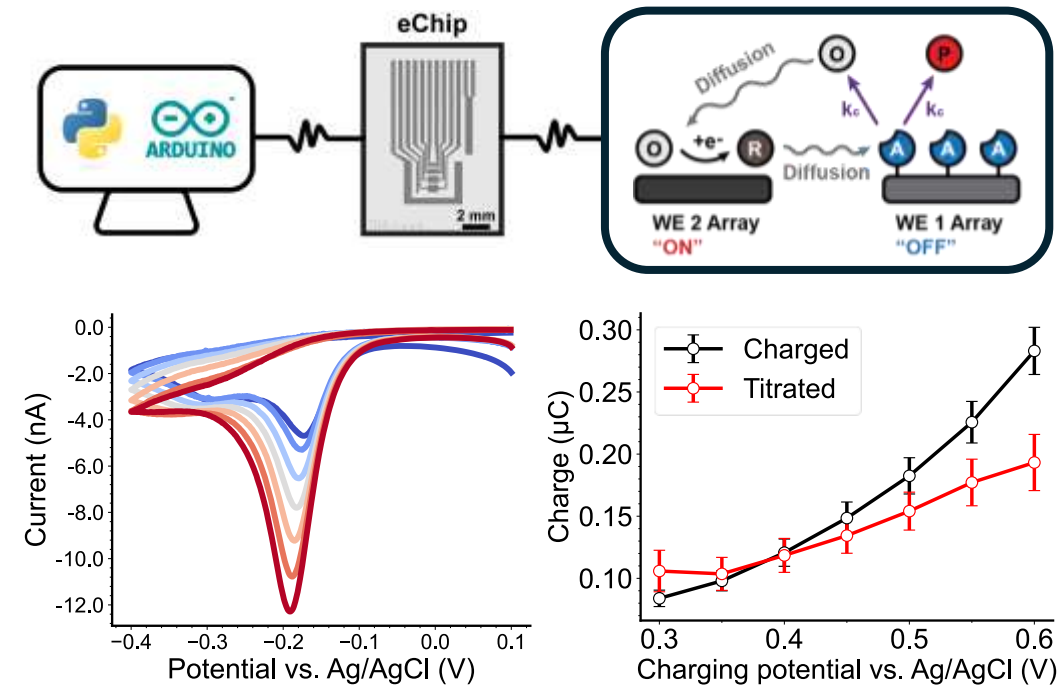
- **500 automated redox titration within 50 hours (2016 CVs, 504 amperometries)**
- 4 IDA gap widths, 7 polypyrrole charging potentials, 3 CV scan rates, and 2 ionic strengths

Conclusion



IDA is a useful platform to accelerate studies of mediated electron transfer processes for energy storage materials

- Developed a workflow to perform automated redox-titration measurements with IDAs to study the charge storage behavior of polypyrrole
- Applied COMSOL simulations understand the kinetics of the redox-titration process
- Revealed the effect of ionic strength to the charge storage capacity of polypyrrole
- Our workflow drastically improves the throughput of redox titration measurements



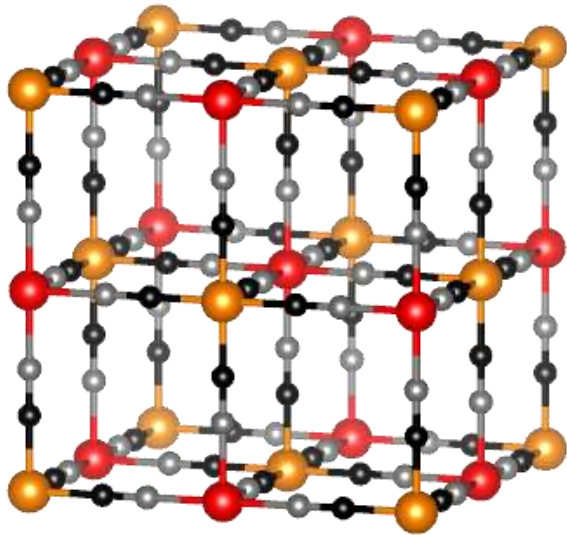
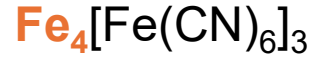
Wang, Z; Pence, M; Rodríguez-López, J.
(Manuscript in preparation)

Studying Prussian Blue with IDAs

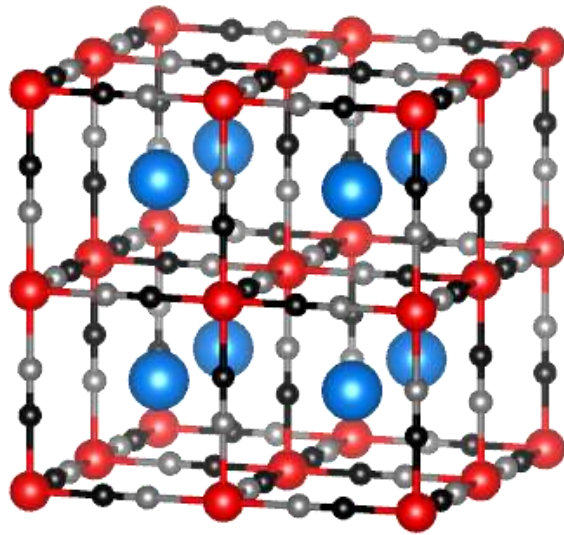
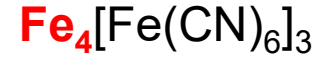


● Fe^{3+} ● Fe^{2+} ○ C ● N ● Cation

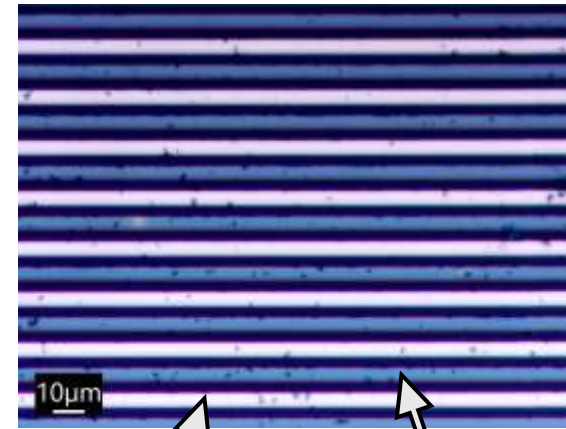
Prussian Blue (PB)



Prussian White (PW)



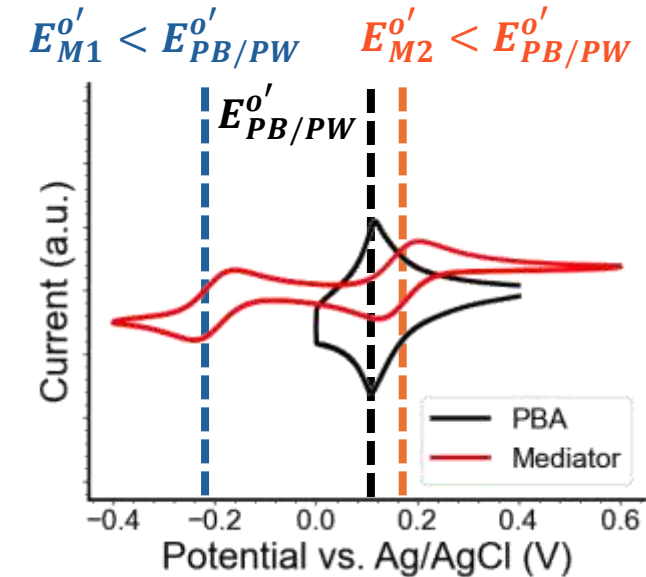
Optical micrograph
after PB deposition



Bare Pt
(Array 2)

PB thin film
(Array 1)

Dual-mediator system



Summary

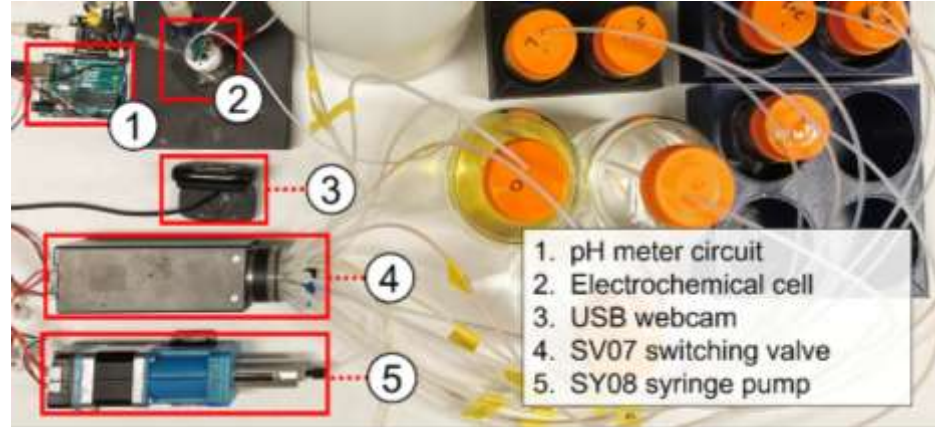
- PB can be electrodeposited on the IDAs
- We began with a dual-mediator system to study the performance of PB

- Prussian Blue (PB) and its analogues are promising materials for RTFBs
- Optimizing their performances requires screening different redox mediators and electrolyte compositions
- **Could the optimization process be accelerated with automated electrochemistry on IDAs?**

Automated Redox Targeting of Prussian Blue



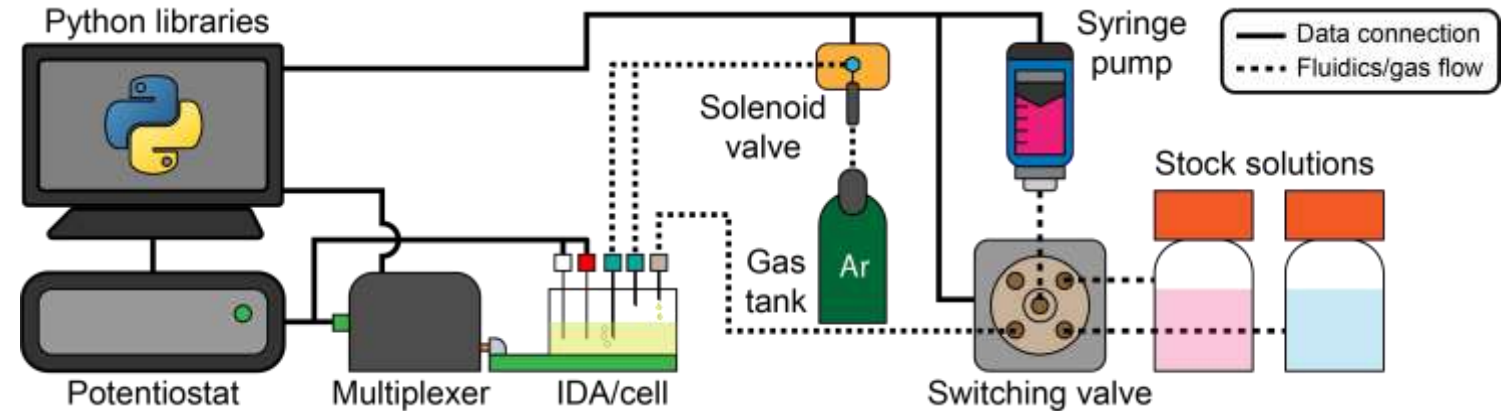
“elab”: A solution handling robot



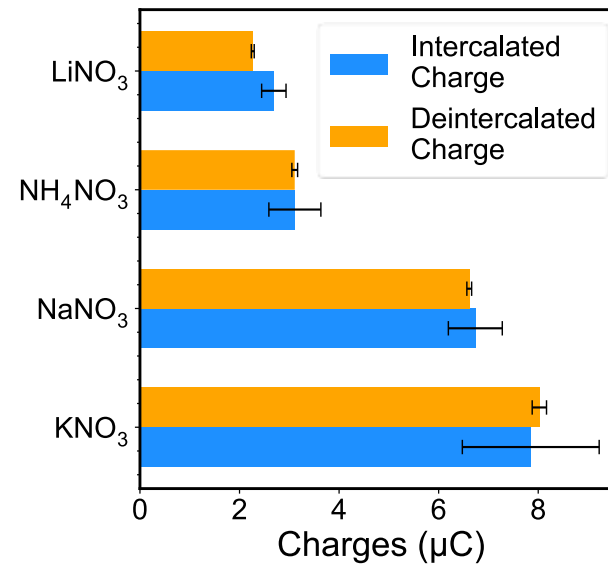
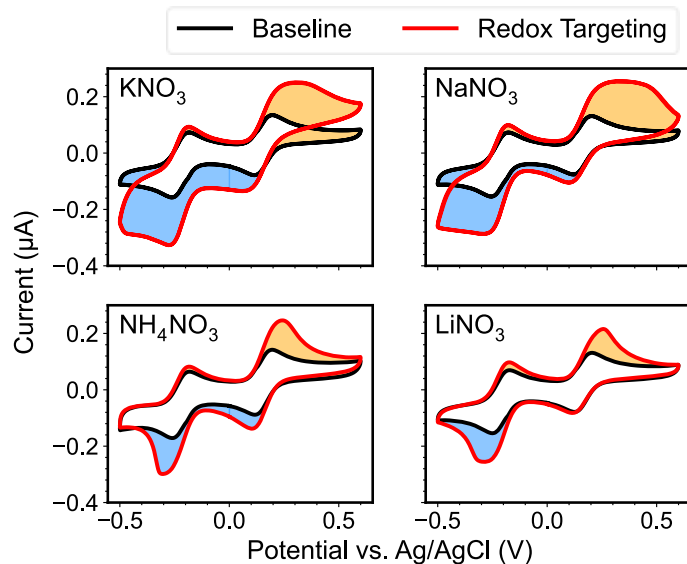
Electrochemistry

Inert gas flow

Solution handling



Automated redox targeting with **different cations**



Summary

- Automated redox-targeting can be performed by coupling IDAs with solution handling hardware.
- It is now possible to systematically screen a large chemical space:** Redox mediators, electrolyte compositions, other Prussian Blue analogues, etc.

Acknowledgements



The JRL lab:

- Prof. Joaquín Rodríguez-López
- Dr. Michael Pence,
- Dr. Oliver Rodríguez, Dr. Abhiroop Mishra
- Gavin Hazen, Raghuram Gaddam, and Seth Putnam
- All the 3rd years in the JRL lab
- **All the JRL lab members!**

Collaborators:

Pacific Northwest National Laboratory

- Dr. Eliot Woods and Dr. Ruozhu Feng

UIUC Cleanroom Research Engineers:

- Joseph Maduzia, Dr. Ranveer Singh
- Dr. Glennys Mensing

L08-2765 – May 22nd (Thursday) 8:30 – 9:10 AM 512D,
presented by Prof. Joaquín Rodríguez-López



The JRL lab, Fall 2024



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Thank you! Questions?



Learn more about
our research!



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